Stick Lengths

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1074 | Sorting and Searching | https://cses.fi/problemset/task/1074

There are n sticks with some lengths. Your task is to modify the sticks so that each stick has the same length.

You can either lengthen and shorten each stick. Both operations cost x where x is the difference between the new and original length.

What is the minimum total cost?

Input

The first input line contains an integer n: the number of sticks.

Then there are n integers: $p_1,p_2,\ldots,p_n$: the lengths of the sticks.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le p_i \le 10^9$

Example

Input
5 2 3 1 5 2

Output
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